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EVAC-1 Software Basic User Guide

1. What is EVAC-1?

EVAC-1 is a powerful AI + Data driven tool designed to help plan for evacuations. It's a web-based application, accessible from any device with an internet connection.

What can EVAC-1 do:

- **Simulate Evacuations:** Model how people might move during an emergency.
- **Identify Bottlenecks:** Find potential traffic jams or other problems.
- **Optimize Evacuation Routes:** Plan the best ways to get people to safety.

2. Getting Started: A Basic Simulation

1. **Accessing the Platform:**
   Login using your assigned email and password.

2. **Name the Simulation:**
   Give the simulation a descriptive name (e.g., "Truckee Wildfire Evacuation + MM/DD/YYYY") to enable easier finding of it later.

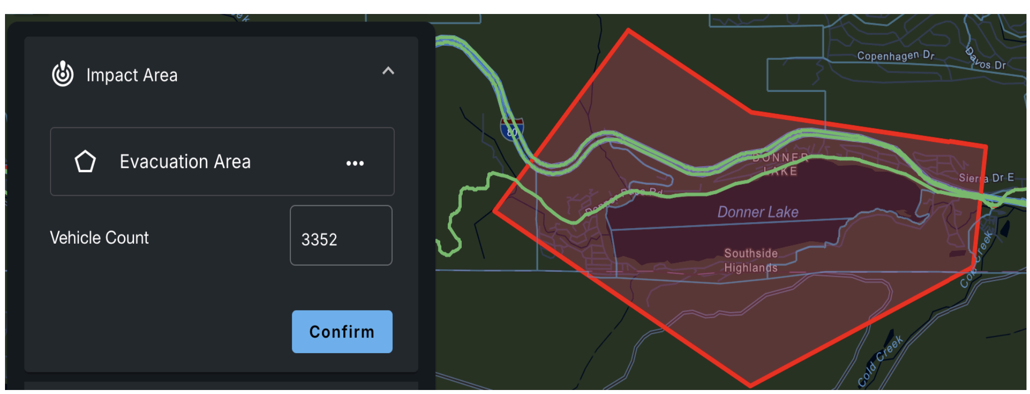
3. **Set Up Traffic Conditions:**
   Choose between:
   - Average Traffic: Use average traffic conditions for all road ways.
   - Live Traffic: Simulates real-time traffic conditions (Note: Live traffic is typically current as of 15 minutes from the first time page is accessed. To have the most current data, the page must be refreshed each time Live Traffic is selected).

3: Defining the Evacuation Scenario

1. Identify the Impact Area (area to be evacuated):

The user can select one of 3 tools to define the impact area:

- **Circle:** Choose a circular area with a specific radius (from 0.1 miles to 5 miles).
- **Zones:** Select predefined zones in your region if they have been loaded in as a dataset.
- **Polygon:** Draw a custom shape to match a specific area (as shown above).



2. Estimating the Vehicle Count (number of vehicles that will evacuate):

- EVAC-1 will provide an automatic estimate of the number of vehicles within the impact area.
- The user can adjust this number to account for factors like day vs night, tourist seasons, or other special events.

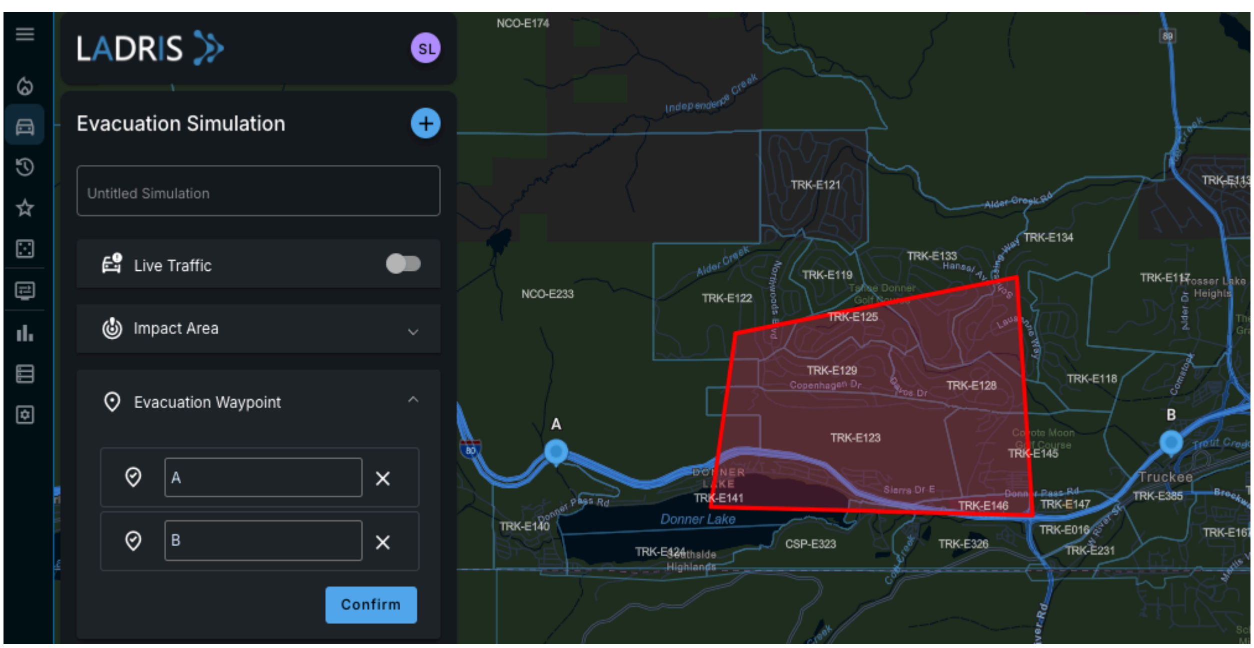
Click “confirm” to move to the next section.

3. Setting Evacuation Waypoints (points of safety):

The user may select waypoints to send evacuated vehicles to in a simulation. Waypoints can be considered a location of safety for evacuating vehicles. The user clicks on the map where they would like to place a waypoint and multiple waypoints can be selected.

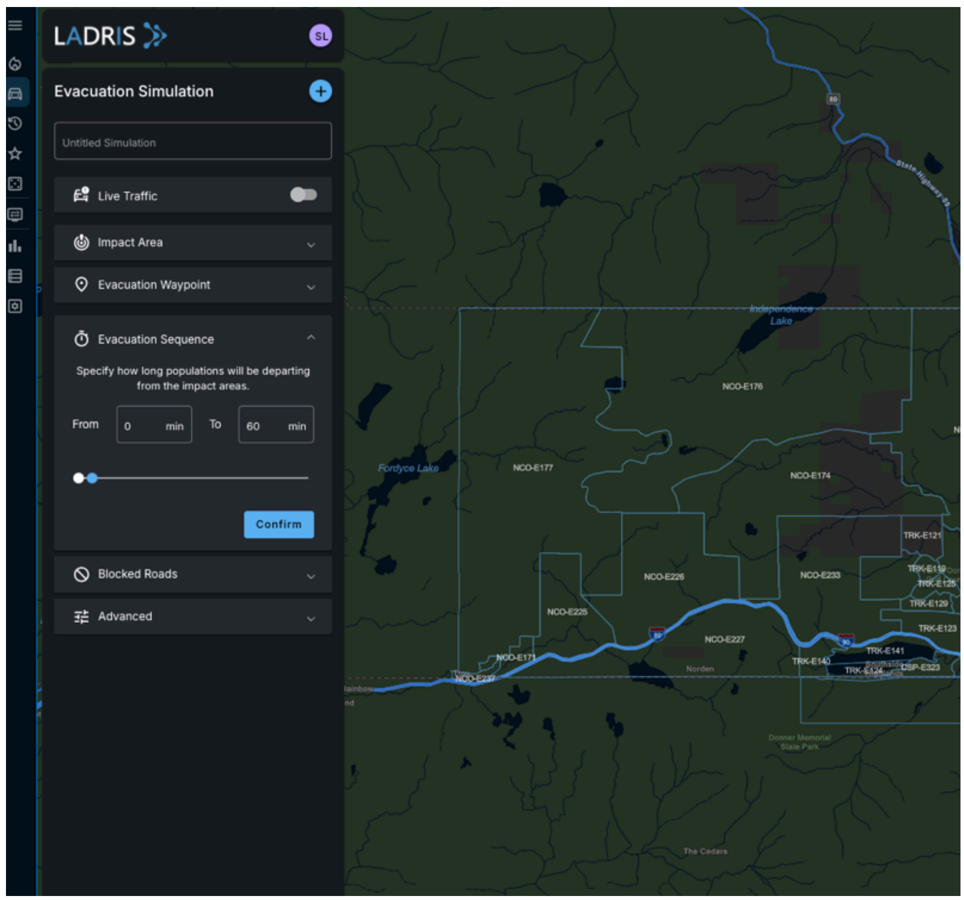
Note: **waypoints are optional** but can be useful in directing the parameters of an evacuation plan. If using waypoints, here are some considerations:

- **Choose Destinations:** Select specific locations where people will evacuate to, such as major roads or large parking areas.
- **Be Precise:** Ensure waypoints are on actual roads or destinations, not just areas on the map that evacuees couldn’t reasonably get to.
- **Consider Travel Time:** Remember evacuation times will include the total time to the waypoint.



4: Setting the Evacuation Sequence

The Evacuation Sequence determines how quickly vehicles within the impact area begin their evacuation. When considering these times, realize when an evacuation alert is issued not everyone will react immediately. Some people may leave right away, while others might take longer to prepare and leave. These times are best determined by the user based on local conditions such as the type of populations, their familiarity with the dangers, and their level of preparedness. By carefully setting the evacuation sequence, the user can create a more accurate simulation of real-world conditions.



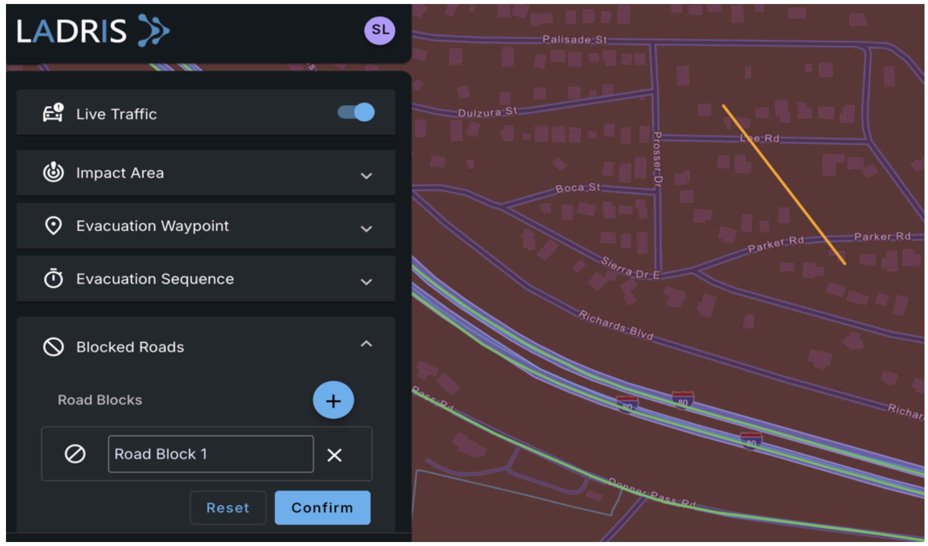
Setting the Sequence:

1. **Define the Time Range:** Set a minimum and maximum time for vehicles to start their evacuation.
   - For example, a user might set a range of 10 minutes to 2 hours. This directly affects total evacuation time, so it's best to select a range that's realistic with the region's estimates or previous history.
2. **Consider Realistic Factors:**
   - **Alert Time:** How long does it take for alerts to reach everyone?
   - **Preparation Time:** How much time do people need to prepare and leave?
   - **Vehicle Availability and Condition:** Are there enough vehicles and are they in good working order?
   - **Population Demographics:** Are there elderly or disabled individuals who might need extra time?

5: Simulating Road Blockages

Blocked roads can significantly impact evacuation times and routes. These blockages can be caused by the incident itself, or used by first responders to control traffic flow.

Creating Road Blockages:



1. **Identify Blocked Roads:** Determine which roads, if any, are impassable due to factors like:
    - Accidents
    - Construction
    - Natural obstacles (e.g., fallen trees, landslides)
    - Emergency response activities
2. **Mark Blockages on the Map:** Use the tool to click and draw lines across the blocked roads.
3. **Multiple Blocked Roads:** A user can create as many blocked roads as they like, either with the **Plus** button or by pressing enter after clicking across a road you'd like to block off and clicking next to another road you'd like to close. Creating a block across multiple roads will shut all roads crossed by the artificial block.

6: Running the Simulation

Once the user has defined the impact area, evacuation waypoints, evacuation sequence, and any blocked roads, click **Run Simulation** at the bottom of the left pane to start the process.

The simulation will visualize the movement of vehicles, identify potential bottlenecks, and estimate evacuation times.

Understanding the Simulation Results:

As the simulation runs, the user sees a visual representation of the evacuation process.

- **Color-Coded Map:**

    Purple Dots: Represents addresses of evacuees.

    Yellow Dots: Indicates partial departures.

    Green Dots: Shows completed departures.

    Green Roads: Shows free-flowing traffic.

    Yellow Roads: Shows moderate traffic.

    Red Roads: Shows heavy traffic and congestion.

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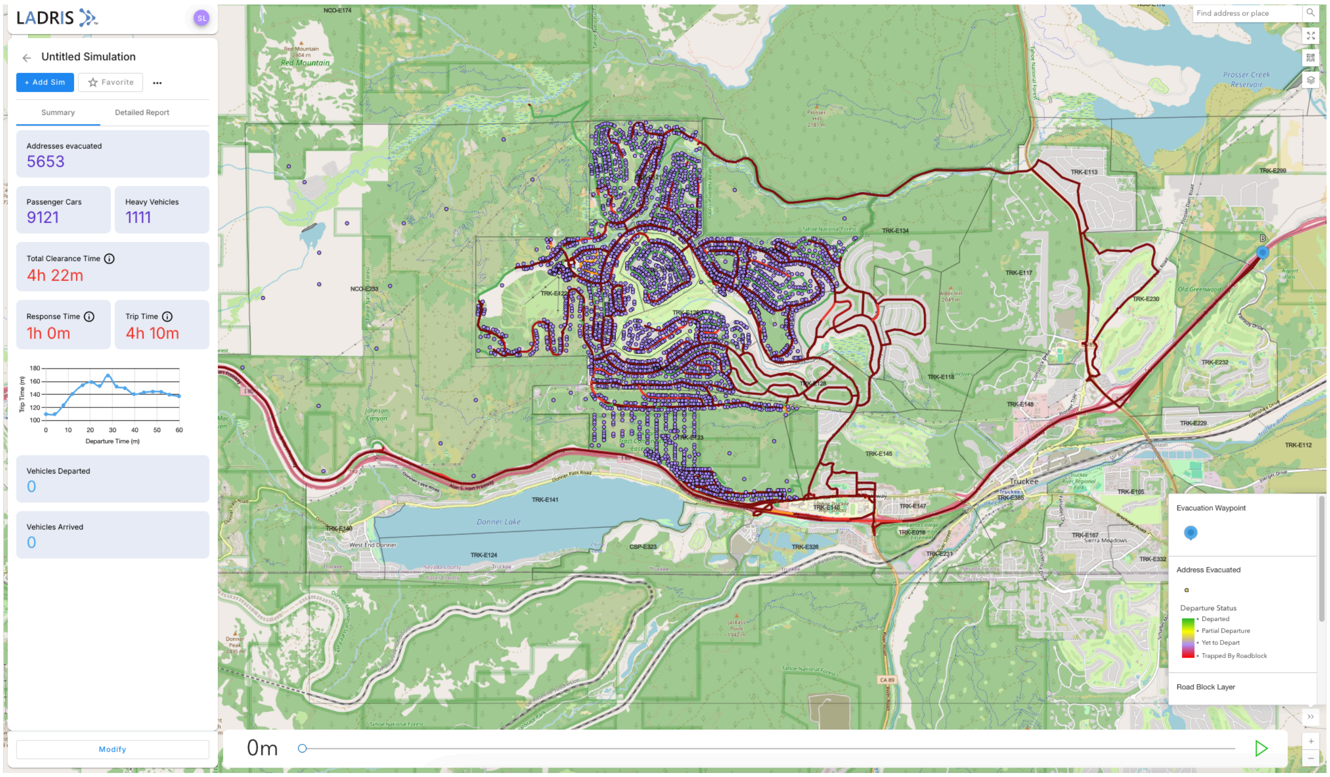
Key Metrics:

- **Total Clearance Time:** The total time it takes for all vehicles to evacuate.
- **Trip Time:** The longest travel time for any individual vehicle.
- **Response Time:** The time between the first and last vehicle departure.



Using Simulation Insights:

By analyzing the simulation results, the user can:

- **Identify Bottlenecks:** Pinpoint areas where traffic congestion is likely to occur.
- **Optimize Evacuation Routes:** Plan more efficient routes to minimize travel times.
- **Improve Emergency Response:** Coordinate emergency services and resource allocation.
- **Educate the Public:** Communicate evacuation plans and potential challenges to residents.

Advanced Simulation Features: For more complex scenarios, EVAC-1 offers advanced features such as:

- **Lane Configuration:** Adjust the number of lanes on roads.
- **Traffic Flow:** Control the direction of traffic (e.g., contraflow).
- **One-Way Streets:** Define one-way roads.
- **Custom Traffic Rules:** Implement specific traffic regulations.

For further assistance and detailed information, please contact the Customer Success team at Ladris.